

PM Accelerator Mission

By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

Weather Trend Forecasting Report

Introduction

Weather Trend Forecasting is to forecast weather conditions based on different previous information such as temperature, air quality, sunrise, and sunset. Before training models, I needed to transform the weather dataset into the modeling weather dataset, so models could be able to learn from the modeling dataset and predict future weather conditions. I utilized Random Forest, XGBoost, and stacking methods to forecast weather conditions. Among these three algorithms, the Random Forest performs the best with a 57% accuracy.

Methodologies

Data Cleaning

Missing values

No missing values are found, I don't need to think how I could handle missing values.

Outliers

I used the winsorization method to deal with outliers. I replaced bottom 5% with its nearest bottom 5% value and replaced top 5% with its nearest top 5% value.

Note: Handling outliers is a difficult task to do in weather forecasting because we know that extreme factors lead to extreme weather conditions. If we remove or replace them with only mean or median, predictive models lose the ability of predicting extreme weather conditions. Moreover, handling outliers in this dataset should be location-specific. For example, wind speeds can vary significantly by region, so an outlier like 100 mph in Kabul should be corrected by referencing the wind data from the last 10 days in the same location. When

handling outliers, it's important to consider only the data corresponding to the specific location, ensuring that the correction is contextually appropriate for that region.

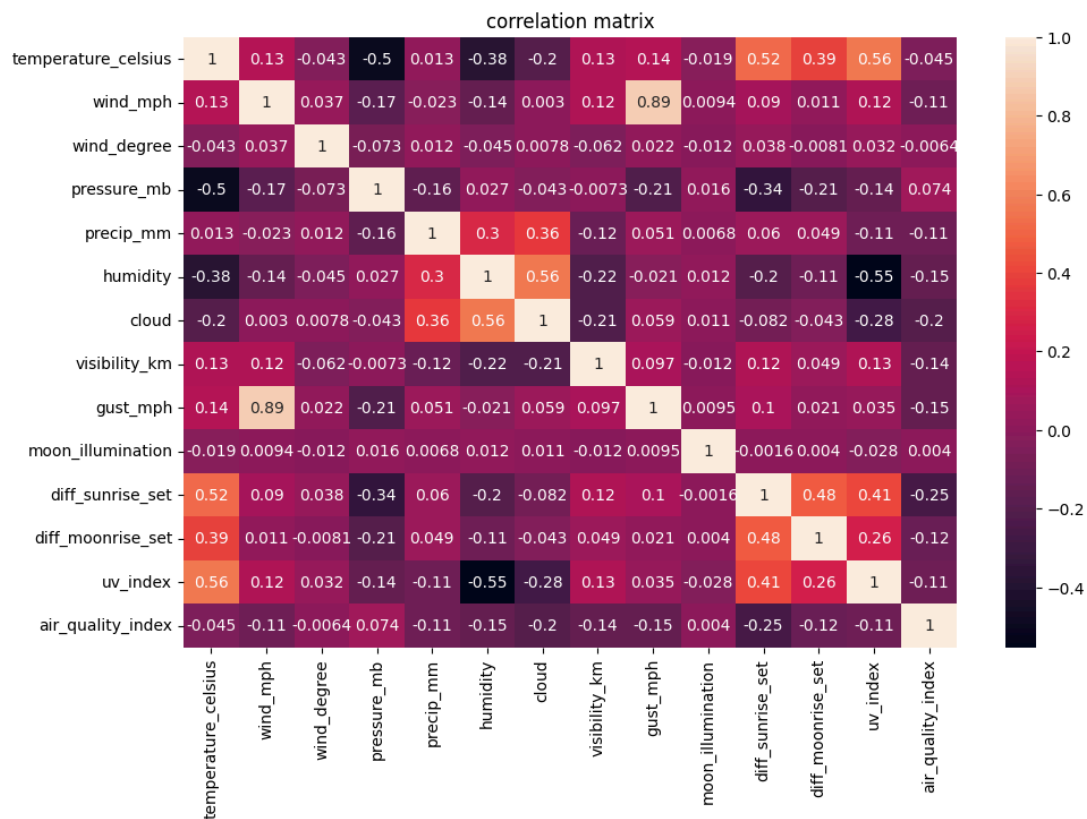
Normalization

I imported StandardScaler() from scikit-learn to standardize the numerical features which has 0 variance and 1 mean.

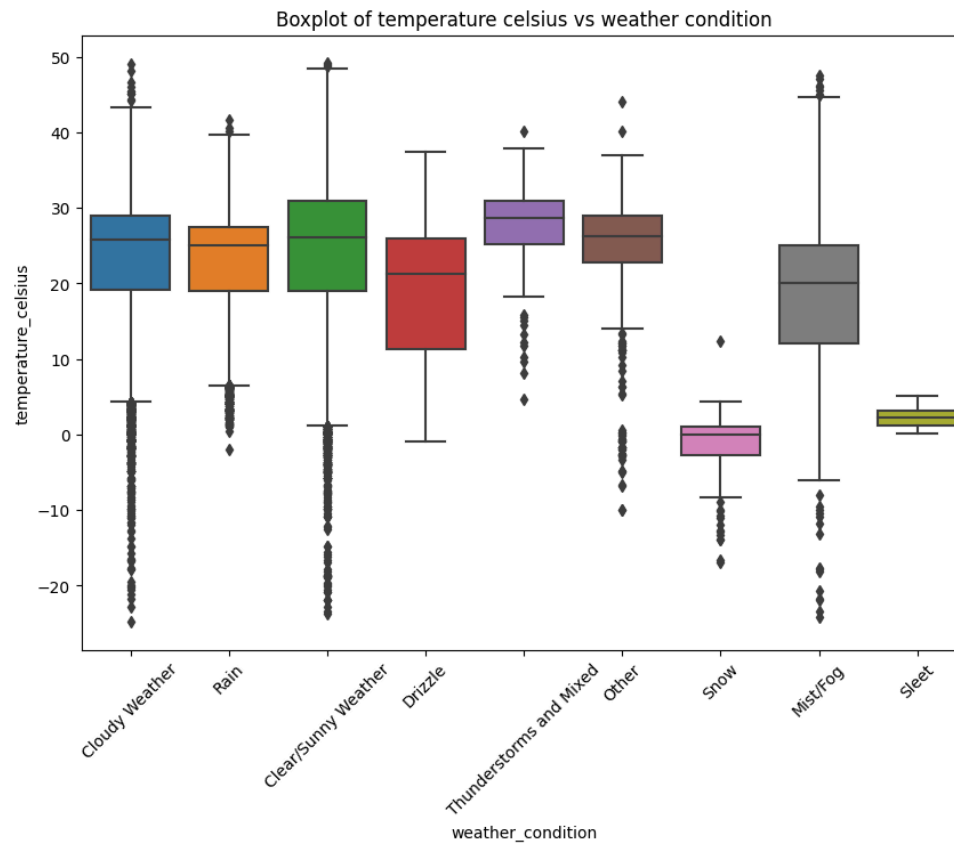
EDA

Correlation

Through a correlation matrix, I can see Most of variables are not correlated. Starting from temperature, temperature is correlated with the difference between sunrise and sunset, and uv_index. Humidity is correlated with clouds. The difference between sunrise and sunset is correlated with the difference between moonrise and moonset, and also uv_index.

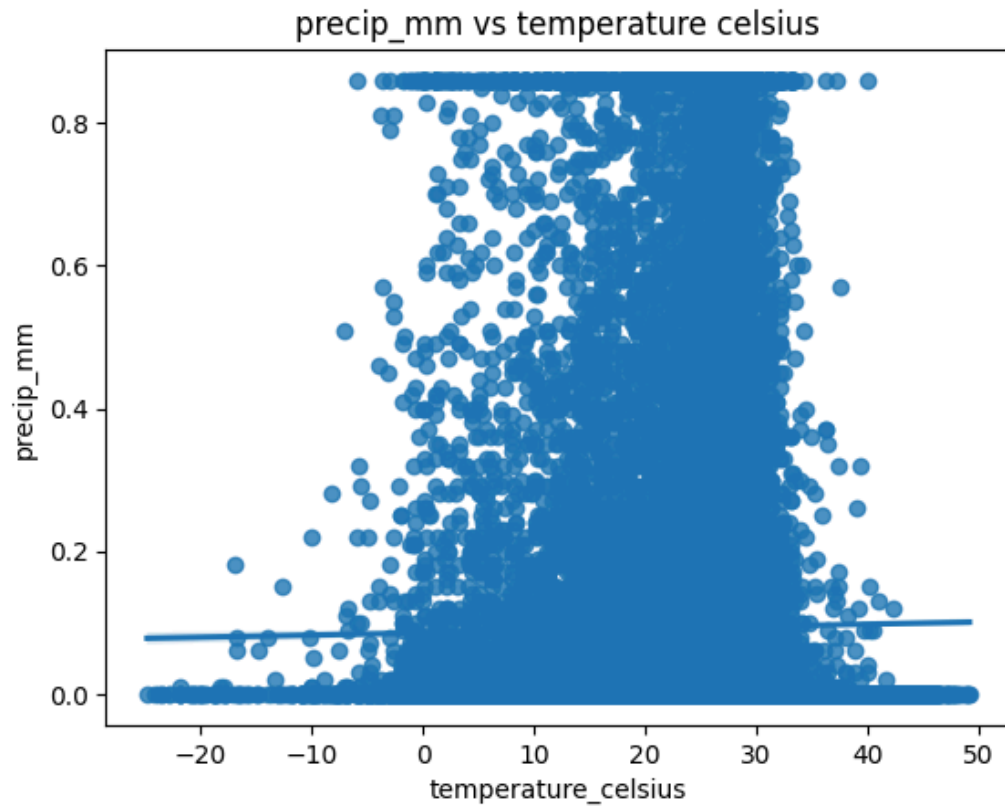


Temperature vs weather condition



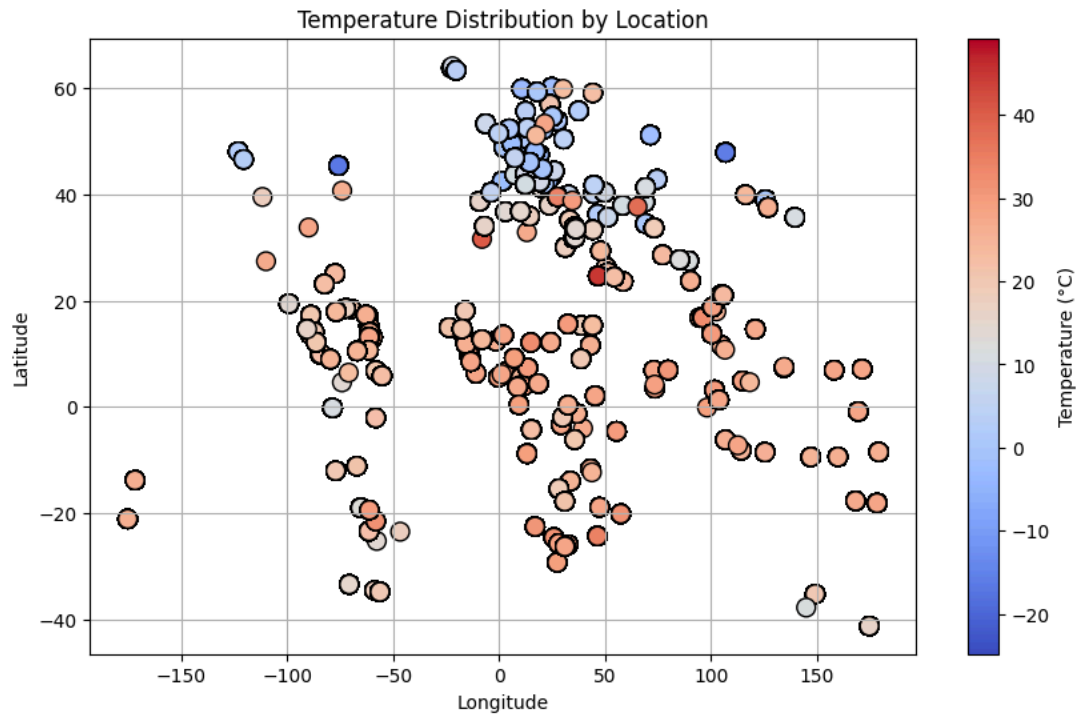
Through box plot temperature vs weather condition, I found the temperature of sunny and cloudy is irregular because they have a lot of negative outliers. This is because of different areas. Some areas are always in a cold temperature.

Temperature VS Precipitation



I plotted the scatterplot of Precipitation vs temperature. We know the correlation between temperature and Precipitation is weak. This may indicate they don't have any connection that higher temperature leads to higher measurement in precipitation

Spatial Analysis



I plotted the scatterplot of latitude vs longitude and color the data point with blue and red.

If a data point is colored in red, it means that the location is under warm weather;

otherwise it is under cold weather. From the plot, as Latitude above 40, we can see there are a lot of places that are under cold weather. I also use `groupby()` to group data by country and calculate the average temperature in each country. 55 countries are under cold weather and 155 countries are under warm weather.

Environmental Impact

I used `groupby()` to group data by weather condition and calculate the average air quality index per weather condition. The freezing drizzle condition has the worst air quality. The Blizzard has the best air quality.

Model Building

Modeling dataset

Since this is a forecasting project, I cannot use current information to predict the label, as the goal is to predict future conditions based on historical data. Each row in the dataset represents the information corresponding to the label, rather than leveraging past information.

To address this, I need to replace the current values with aggregated historical data. For instance, if Kabul recorded 22°C on June 11, 2024, I would calculate the average temperature over the previous 10 days and use that value instead. This approach ensures the model relies on prior information to forecast future weather conditions accurately.

Moreover, we can reduce complexity of the dataset by grouping air quality index and uv index into sub categories. I can group air quality indexes into Low, Moderate, High, and Very High. I can group the uv indexes into No Protection Required, Protection Required, and Extra Protection Required. Since I had a lot of weather conditions to predict, I also grouped them into sub groups as well. I group weather conditions into Rain, Snow, Sleet, Drizzle, Thunderstorm and Mixed, Clear/Sunny Weather, Cloudy Weather, and Mist/Fog.

I used one-hot encoding to transform categorical features into binary numbers.

Models

I splitted data into 70% of training and 30% of testing. I used Random Forest, XGB Classifier, and stacking models for this project. Initially, I runed Random Forest and XGBClassifier with grid search and principal component analysis (PCA) to find the best combination of hyperparameter and PCA component to maximize accuracy, F1-score, and Precision. Then, once I found the best combination of hyperparameter and PCA component, I put these models into a stacking method to try to generalize results with two ensemble models.

Random Forest

	precision	recall	f1-score	support
0	0.63	0.62	0.62	4905
1	0.57	0.14	0.23	1988
2	0.54	0.79	0.64	6271
3	0.62	0.13	0.22	961
4	0.00	0.00	0.00	131
5	0.00	0.00	0.00	132
6	0.00	0.00	0.00	9
7	0.00	0.00	0.00	209
accuracy			0.57	14606
macro avg	0.30	0.21	0.21	14606
weighted avg	0.56	0.57	0.53	14606

From the results of random forest, the accuracy was only 57% The true label accuracy from 0 to 3 was acceptable but from 5 to 7 did not look good because the Random Forest did not even capture trends of these types of weather conditions in the dataset. One possible reason I could think is that the training dataset does not include information from label 5 to 7.

XGB

	precision	recall	f1-score	support
0	0.61	0.63	0.62	4905
1	0.45	0.25	0.32	1988
2	0.56	0.73	0.63	6271
3	0.54	0.13	0.20	961
4	0.00	0.00	0.00	131
5	0.00	0.00	0.00	132
6	0.00	0.00	0.00	9
7	0.00	0.00	0.00	209
accuracy			0.57	14606
macro avg	0.27	0.22	0.22	14606
weighted avg	0.54	0.57	0.54	14606

The XGB almost has the same result as Random Forest through Precision, recall, and accuracy metrics.

Stacking Methods

	precision	recall	f1-score	support
0	0.65	0.61	0.63	4905
1	0.53	0.20	0.29	1988
2	0.55	0.78	0.64	6271
3	0.55	0.13	0.21	961
4	0.00	0.00	0.00	131
5	0.00	0.00	0.00	132
6	0.00	0.00	0.00	9
7	0.00	0.00	0.00	209
accuracy			0.58	14606
macro avg	0.28	0.22	0.22	14606
weighted avg	0.56	0.58	0.54	14606

The stack of Random Forest and XGB cannot improve anything.

Final Thoughts

Improving models require more time in feature engineering or simply getting more observation is a good method to try. Thank you for reading my report!